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Multi-kernel Scale Transformation Enhancement Network with Gated Feature Fusion for Remote Sensing Object Detection

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***Abstract*—Remote sensing object detection generally faces challenges such as large scale variations and severe background interference. Existing methods mainly improve detection performance by enlarging the receptive field or enhancing contextual dependencies. However, the former tends to weaken the texture information of small objects, while the latter is often constrained by limited local context, resulting in insufficient feature representation. To address these issues, we propose a Multi-kernel Scale Transformation Enhancement Network(MSTENet). Specifically, Scale Transformation Module reconstructs multi-scale receptive fields to capture objects of varying scales while modeling long-range contextual information. To alleviate the loss of small-object features caused by multi-scale receptive field modeling, Fine-grained Enhancement Module is cascaded after STM to compensate for the fine-grained information of small objects that may be lost during scale transformation while enhancing local contextual representation. Furthermore, to balance the collaborative effects of STM and FEM, we introduce a Gated Feature Fusion mechanism that adaptively integrates the global features from STM with the local features from FEM, thereby enhancing foreground–background discrimination and category-boundary representation. Experimental results show that MSTENet achieves mAP scores of 82.72%, 90.67%, and 68.84% on the DOTA-v1.0, HRSC2016, and DIOR datasets, respectively, demonstrating the effectiveness of the proposed method.**

***Index Terms*—Remote sensing, Multiscale variation, Context aggregation, Feature fusion.**

I. Introduction

With the rapid development of observation technologies, remote sensing data have been widely applied in various fields, including military applications, transportation, and agriculture. Consequently, object detection in remote sensing imagery has attracted considerable attention and has become an active research topic in recent years [1], [2]. Unlike conventional object detection methods [3]–[5], which only provide coarse object localization using horizontal bounding boxes, oriented object detection in remote sensing imagery aims to accurately determine object orientations and generate precise Oriented Bounding Boxes (OBBs). However, remote sensing images are typically captured with a large field of view, often resulting in low spatial resolution and poor contrast [6]. Moreover, challenges such as large-scale variations, weak discriminative features, and ambiguous object orientations [7], [8], [9] further increase the difficulty of accurate detection. Therefore, effectively addressing these challenges has become an important research direction in the field of remote sensing object detection.

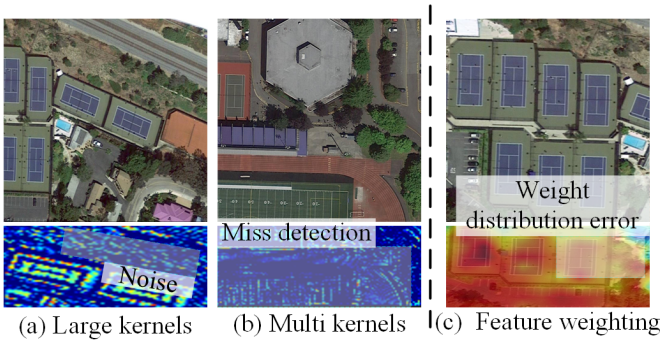

Fig. 1. Typical limitations of existing remote sensing object detection methods. (a) Large-kernel convolutional networks may treat small objects as noise and thus suppress their responses. (b) Multi-scale convolutional networks may cause the loss of target information when handling objects with high aspect ratios. (c) Traditional feature-weighting mechanisms may mistakenly assign excessive attention to background regions.

Existing studies have mainly focused on improving the angle prediction of OBBs [10], [11], [12] and designing high-performance OBB detectors [13], [14], [15]. Although these methods have achieved considerable progress, several limitations remain, as illustrated in Fig. 1. In terms of receptive field modeling, LSKNet relies on large-kernel convolutions to capture long-range contextual information. However, in scenes containing small objects or complex backgrounds, this strategy tends to introduce excessive irrelevant information, thereby weakening the fine-grained features of small objects. Although PKINet improves scale adaptability through multi-scale convolutions, it remains insufficient in modeling the directional structures of objects with large aspect ratios, such as bridges and harbors. Meanwhile, at the feature representation level, when objects and backgrounds exhibit similar characteristics in terms of scale, color, and texture, existing methods often struggle to accurately distinguish their relative importance. This can lead to imbalanced feature weighting, where background responses are excessively enhanced while

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effective features in target regions are suppressed.

To address the these issues, we propose an efficient detection framework, termed MSTENet. Unlike conventional multi-scale methods, MSTENet rethinks the design of network receptive fields. Specifically, the Scale Transformation Module (STM) combines square convolution kernels for enlarging the receptive field with elongated convolution kernels for capturing objects with large aspect ratios, and constructs multiple branches with different kernel sizes to extract both global contextual information and fine-grained texture features. To alleviate the loss of small-object information caused by excessively large receptive fields, a Fine-grained Enhancement Module (FEM) is cascaded after STM to strengthen local contextual information. Furthermore, the two modules are integrated through a Gated Feature Fusion (GFF) mechanism, which dynamically fuses global and local features, thereby enabling more effective discrimination between foreground and background information.

Overall, the main contributions of this work are summarized as follows:

- To address the issue of scale variation in remote sensing object detection, we design a Scale Transformation Module (STM), which enlarges the receptive field while more effectively capturing elongated objects. Meanwhile, to alleviate the loss of small-object features caused by receptive field expansion, a Fine-Grained Enhancement Module (FEM) is cascaded after STM to enhance the representation of small-object features and local information.
- To facilitate effective collaboration between STM and FEM, we propose a Gated Feature Fusion (GFF) mechanism to dynamically integrate the global information extracted by STM and the local information enhanced by FEM, thereby improving foreground-background discrimination and the representation of inter-class boundaries.
- Extensive experiments on the DOTA-v1.0, HRSC2016, and DIOR datasets demonstrate that MSTENet achieves state-of-the-art detection performance and consistently outperforms existing oriented object detectors.

II. RELATED WORK

A. Remote Sensing Object Detection

Existing remote sensing object detection methods mainly improve detection performance from two aspects: detector architecture design and oriented object modeling. In terms of detector architecture, AOPG [15] adopts an anchor-free strategy to generate coarse oriented proposals and further refines them, while R3Det [16] improves the localization accuracy of oriented objects through progressive regression and feature refinement. LSKNet [17] and PKINet [18] employ large convolution kernels to dynamically adjust the receptive field and multi-scale convolutions to extract features of objects at different scales, respectively. In terms of oriented object modeling, RoI Transformer [19] achieves object-region alignment by learning the spatial transformation from horizontal RoIs to rotated RoIs, while ReDet [10] models orientation variations using rotation-equivariant networks and Rotation-invariant RoI Align. Oriented RepPoints [20] describes the geometric structures of arbitrarily oriented objects using adaptive point sets. S^2ANet [21] employs Alignment Convolution to adjust feature sampling locations according to rotated anchors, thereby aligning convolutional features with the spatial geometry of objects. ARC [22] further enhances feature extraction for objects with different orientations through adaptive rotation convolution kernels. Oriented R-CNN [23] adopts a six-parameter midpoint representation and directly generates oriented proposals using an Oriented RPN. PSC [24] addresses the periodicity of object orientations through phase encoding, while related angle optimization methods further improve orientation prediction by alleviating the boundary discontinuity problem in rotated bounding box regression.

B. Remote sensing target feature fusion

In remote sensing imagery, scale variations, complex backgrounds, and weak object responses make multi-scale feature fusion and contextual information modeling important strategies for enhancing object representation. Early methods mainly focused on feature interactions across different levels. SCRDet [25] employs a Sampling Fusion Network to fuse features from different layers and combines pixel attention and channel attention to enhance object responses. CAD-Net [26] models global and local contextual information at the scene and object levels, respectively, and utilizes spatial-scale-aware attention to select effective regions and appropriate feature scales. AugFPN [27] improves multi-scale feature fusion across different levels through consistency supervision, residual feature augmentation, and adaptive RoI selection, while HRDNet [28] jointly employs a multi-depth image pyramid and a multi-scale feature pyramid to preserve high-resolution details and deep semantic information. SCANet [29] extracts multi-scale information through branches with different receptive fields and progressively integrates high-level semantic context into low-level features, whereas SCBi-FPN [30] strengthens cross-scale information propagation through a scene-context-aware bidirectional feature pyramid. More recently, TMAFNet [31] enhances object representation in complex backgrounds through task-balanced multi-scale adaptive fusion. AGMF-Net [32] simultaneously models long-range dependencies and multi-scale object features via global information extraction and multi-scale feature fusion. LR-FPN [33] further extracts positional and saliency information from shallow features and injects them into different pyramid levels through spatial and channel interactions. In addition, PC-RCNN [34] adopts parallel feature fusion and cascaded global attention to strengthen cross-level information interaction, whereas BAFNet [35] jointly models objects and their surrounding backgrounds through dual-stream attention and leverages boundary information to enhance the detailed representation of small objects in remote sensing imagery.

III. METHOD

MSTENet consists of three key modules, namely STM, FEM, and GFF, as illustrated in Fig. 2. STM employs multi-scale receptive fields to capture objects at different scales

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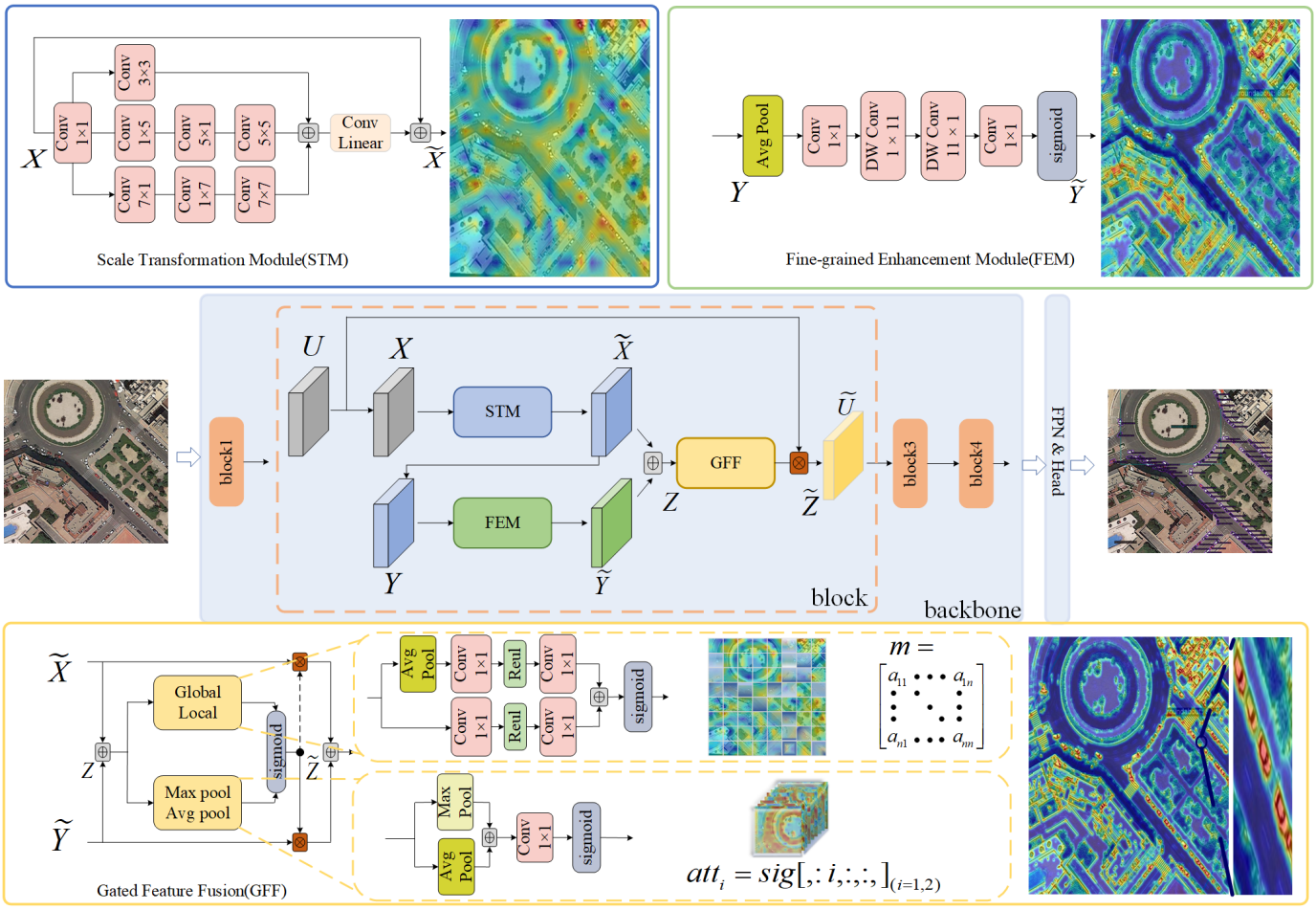

Fig. 2. Overview of MSTENet. MSTENet consists of four stages, each of which contains a Feed-Forward Network (FFN) composed of STM, FEM, and GFF. STM extracts target features through multi-scale and multi-shape receptive fields while modeling global contextual information. FEM further compensates for the fine-grained information of small and elongated targets and enhances local representations. GFF generates a fusion coefficient m and spatial attention maps att_i based on the features from the two branches. Specifically, m is used to balance the contributions of STM and FEM, while att_i reweights the spatial responses of each branch. By dynamically integrating global and local features, GFF ultimately enhances target representations and improves foreground–background discrimination.

and model global contextual information. Subsequently, FEM compensates for the fine-grained information of small objects that may be lost during scale transformations while enhancing local contextual representations. Finally, GFF jointly exploits the global features extracted by STM and the local features enhanced by FEM, and adaptively integrates them in a mix-up manner, thereby improving foreground–background discrimination.

A. Scale Transformation Module

To address the limitations of large-kernel convolutions and multi-scale convolutions in background suppression and elongated-target modeling, we revisit the effects of convolution kernel scale and shape on receptive-field modeling and propose STM. By employing convolution kernels with different sizes and shapes to construct multi-scale receptive fields, STM enhances the feature extraction capability for targets with different scales and orientations. STM consists of three branches. First, 1×1 convolution is applied to the input feature X for channel mapping:

$$L = \mathbb{F}_{\mathrm{Conv}}^{1\times 1}(X) \tag{1}$$

where $L \in \mathbb{R}^{C\times H\times W}$ is used to adjust the channel dimension to accommodate the subsequent three branches. Then, the convolutional combination function C_n is defined as:

$$\mathcal{C}_n(X) = \begin{cases} \mathbb{F}_{\mathrm{Conv}}^{3\times 3}(L) & n=1 \\ \mathbb{F}_{\mathrm{Conv}}^{5\times 5}\left(\mathbb{F}_{\mathrm{Conv}}^{5\times 1}\left(\mathbb{F}_{\mathrm{Conv}}^{1\times 5}(L)\right)\right) & n=2 \\ \mathbb{F}_{\mathrm{Conv}}^{7\times 7}\left(\mathbb{F}_{\mathrm{Conv}}^{1\times 7}\left(\mathbb{F}_{\mathrm{Conv}}^{7\times 1}(L)\right)\right) & n=3 \end{cases} \tag{2}$$

Specifically, branch $n = 1$ employs a 3×3 convolution to extract local texture and edge information, providing basic feature representations for targets at different scales. Branches $n = 2$ and $n = 3$ further introduce strip convolutions with different kernel sizes. By aggregating features along the horizontal and vertical directions, these branches enlarge the receptive field and enhance the perception of directional structures and edge textures of elongated targets, such as ships and bridges. The different convolution kernel sizes and arrangements enable the three branches to complementarily model target features from different scales and orientations.

Subsequently, the outputs of the three branches are aggregated, and multi-branch feature fusion is performed through

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linear mapping. A residual connection is established between the fused feature and the input X. After a 1×1 convolution and a ReLU activation function, the scale-transformed feature $\widetilde{X}$ is obtained:

$$\widetilde{X} = \mathrm{ReLU}\left(\mathbb{F}_{\mathrm{Conv}}^{1\times1}\left(X + \mathbb{F}_{\mathrm{ConvLinear}}\left(\sum_{n=1}^{3}\mathcal{C}_n(X)\right)\right)\right) \quad (3)$$

$$Y = \widetilde{X} \quad (4)$$

where $\mathbb{F}_{\mathrm{Convlinear}}$ denotes linear feature mapping, which is used to aggregate the multi-scale and multi-directional information extracted by the three branches and form a unified feature representation. The residual connection preserves the original features, facilitates information propagation, and alleviates the vanishing-gradient problem in deep networks. Finally, the resulting X is used as the input for the subsequent processing.

B. Fine-grained Enhancement Module

It is observed that STM mainly aggregates target features at different scales and global contextual information through multi-scale receptive fields, while its capability for further modeling local fine-grained information remains limited. To address this issue, FEM is cascaded after STM to further enhance target textures, edges, and structural continuity through local feature refinement and directional structure modeling. Similarly, the convolutional combination in the fine-grained enhancement process is denoted by C, which is formulated as:

$$\mathcal{C}(Y) = \mathbb{F}_{\mathrm{Conv}}^{1\times1}\left(\mathbb{F}_{\mathrm{DWConv}}^{11\times1}\left(\mathbb{F}_{\mathrm{DWConv}}^{1\times11}\left(\mathbb{F}_{\mathrm{Conv}}^{1\times1}(Y)\right)\right)\right) \quad (5)$$

Here, $\mathbb{F}_{\mathrm{DWConv}}$ denotes depthwise convolution. FEM is constructed by cascading standard convolutions and depthwise strip convolutions. Specifically, the 1×1 convolution is employed for local feature mapping and channel-wise information interaction, while the 1×11 and 11×1 depthwise strip convolutions aggregate features along the horizontal and vertical directions, respectively. In this way, directional texture and structural information can be enhanced while maintaining a relatively low computational overhead. This structure is particularly suitable for elongated targets, such as bridges and ships, as it strengthens edge responses along the principal axis of the target while preserving structural continuity. Subsequently, average pooling is employed to aggregate local neighborhood information, and the convolutional combination together with a Sigmoid function is used to generate the fine-grained enhanced feature map:

$$\widetilde{Y} = Y \cdot \mathrm{sig}\left(C\left(P_{\mathrm{avg}}(Y)\right)\right) \quad (6)$$

Here, P_{avg} denotes average pooling, and sig denotes the Sigmoid function, which constrains the feature weights to the range $(0, 1)$, thereby adaptively enhancing effective local texture and structural information. For small targets, discriminative information is typically concentrated in weak textures, fine edges, and locally continuous regions, which can be easily weakened during multi-scale feature extraction.

FEM enhances these target features through local response refinement and neighborhood information aggregation. For elongated targets, the depthwise strip convolutions enlarge the effective receptive field along the horizontal and vertical directions, enabling the network to more comprehensively model directional textures, long edges, and structural continuity. Therefore, FEM not only compensates for the fine-grained information that is easily lost for small targets, but also further enhances the structural representation capability for targets with elongated targets, such as bridges and ships.

C. Gated Feature Fusion

Low contrast in remote sensing images often results in ambiguous foreground–background responses. Conventional fusion strategies, such as direct summation or concatenation, are insufficient to adaptively exploit the complementary information provided by different feature sources. To this end, GFF is introduced to dynamically integrate the multi-scale global features from STM and the local fine-grained features from FEM, thereby enhancing discriminative target responses while suppressing background interference. The two feature streams are first aggregated as:

$$Z = \widetilde{X} + \widetilde{Y} \quad (7)$$

Here, $\widetilde{X}$ denotes the multi-scale global features output by STM, while $\widetilde{Y}$ denotes the local fine-grained features output by FEM. To fully exploit the information contained in these two types of features, GFF performs adaptive fusion from two aspects, namely feature contribution weighting and spatial response features.

At the feature contribution weighting level, GFF models contextual information through a local path and a global path, respectively. Specifically, the local path employs 1×1 convolutions to extract local responses, while the global path combines global average pooling with 1×1 convolutions to encode global contextual information. The two types of contextual information are fused after channel mapping, and a gating coefficient m is generated as:

$$C(Z) = \mathbb{F}_{\mathrm{conv}}^{1\times1}\mathbb{F}_{\mathrm{conv}}^{1\times1} + \mathbb{F}_{\mathrm{conv}}^{1\times1}\mathbb{F}_{\mathrm{conv}}^{1\times1}\left(P_{\mathrm{avg}}(Z)\right) \quad (8)$$

$$m = \mathrm{sig}(C(Z)) \quad (9)$$

Here, m has a shape of $(B, 1, 1, 1)$ and is broadcast to $(B, 1, H, W)$ during computation so that its dimensions are compatible with those of the feature maps.

At the spatial feature level, GFF further generates spatial feature maps corresponding to the two feature streams. Specifically, average pooling and max pooling are performed along the channel dimension to extract complementary spatial statistical information. The two outputs are then concatenated, followed by convolution and an activation function to generate two spatial attention maps:

$$\mathrm{att}_i = \mathrm{sig}\left(P_{\mathrm{avg}}(Z) + P_{\max}(Z)\right)_{(i=1,2)} \quad (10)$$

Here, the weights obtained through the Sigmoid function can be represented as $\mathrm{sig}[:, i, :, :]$. Through broadcasting, the shape of $\mathrm{sig}[:, i, :, :]$ is reshaped to $\mathrm{sig}[B, i, H, W]$, allowing it to

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Algorithm 1 Forward Pass of MSTEblock

Input:
Feature map: $U \in \mathbb{R}^{C\times H\times W}$
Output:
Enhanced feature map: $\widetilde{U} \in \mathbb{R}^{C\times H\times W}$
// Step 1: preprocessing
$att = \text{process}(U)$
// Step 2: Extract multi-scale and fine-grained features
$att_1 =$ via Eq.(3)
$att_2 =$ via Eq.(6)
$att_i = att_1 + att_2$
// Step 3: Generate spatial weighting matrix and attention maps
$m =$ via Eq.(9)
$att_i =$ via Eq.(10)
// Step 4: Mix-up linear combination
$\widetilde{U} = att_1 \cdot m + att_2 \cdot (1 - m)$
return $\widetilde{U}$;

perform element-wise multiplication with the feature maps. The final gated feature fusion can be mathematically expressed as:

$$\widetilde{Z} = \widetilde{X} \cdot \text{att}_1 \cdot m + \widetilde{Y} \cdot \text{att}_2 \cdot (1 - m) \tag{11}$$

It can be observed that when m is relatively large, the model places greater emphasis on the multi-scale features extracted by STM; when m is relatively small, the model relies more on the local fine-grained information obtained by FEM. Through the dynamic adjustment of m, the model can adaptively balance scale-varying information and fine-grained enhanced information, thereby highlighting critical features while suppressing redundant responses.

The final output of the MSTENet block can be regarded as the element-wise multiplication between the input feature U and the weighted feature $\widetilde{Z}$:

$$\widetilde{U} = \widetilde{Z} \cdot U \tag{12}$$

Overall, MSTENet employs STM to model multi-scale targets and global contextual information, utilizes FEM to compensate for local fine-grained and directional structural information, and adopts GFF to adaptively fuse these two types of complementary features, thereby improving target representation and foreground–background discrimination in complex scenes. The complete workflow of the MSTENet block is presented in Algorithm 1.

IV. Experiment

A. Dataset

1) DOTA-v1.0 [55]: is a large-scale object detection dataset for optical remote sensing images, which contains 15 categories, 2806 images of different sizes, and 188,282 individual instances, with image widths and heights ranging from 800 to 4000 pixels. The dataset is divided into training, validation, and testing sets of 1,411, 458, and 937 images respectively. Due to the varying input scales of the original images, all images were cropped into 1024×1024 blocks overlapping by 200 pixels for the purpose of data cleaning and processing.

2) HRSC2016 [56]: is an optical remote sensing image dataset specifically designed for ship detection. It contains 1,680 images and 2,976 instances, with image resolutions ranging from 300×300 to 1500×900. After removing samples without annotations, we obtain 1,061 valid annotated images. We split the dataset into 436 images for training, 181 images for validation, and 444 images for testing. The input size is set to 800×800 while preserving the aspect ratio. During training, horizontal and vertical flipping as well as random rotation are applied to increase the scale and diversity of the training data.

3) DIOR [57]: is a large-scale aerial image dataset for oriented object detection in remote sensing, containing 23,463 images with 192,518 annotated instances across 20 object categories. The dataset is divided into 5,862 training images, 5,863 validation images, and 11,738 testing images. In our experiments, the training and validation sets are combined, resulting in 11,725 images for training, while the remaining 11,738 images are used for testing. The input image size is set to 800×800 pixels, consistent with HRSC2016.

B. Implementation Details

Our model is implemented in PyTorch based on the open-source MMRotate framework. All experiments are trained on eight NVIDIA RTX 4090 GPUs, while evaluation is performed on a single NVIDIA L20 GPU.

For the model architecture, we adopt Oriented R-CNN as the detection framework, with our proposed MCSTNet as the backbone and the official FPN implementation from MMDetection as the neck. The RPN stage employs OrientedRPNHead to generate region proposals, while the RoI stage uses RoIAlignRotated for rotated region feature extraction. Both the classification and regression heads consist of two fully connected layers. For the loss functions, the RPN adopts sigmoid cross-entropy loss for classification and Smooth L1 loss for regression. Similarly, the RCNN stage employs cross-entropy loss for classification and Smooth L1 loss for regression. For optimization, we employ the AdamW optimizer and train the model for 12 epochs. The initial learning rate is set to 2×10^{-4}, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 0.05. A step-based learning rate schedule is adopted, with a linear warmup over the first 500 iterations and a warmup ratio of $1/3$. The learning rate is subsequently decayed at the 9th and 11th epochs. Additionally, gradient clipping with a maximum norm of 35 is applied to improve training stability. For data preprocessing, dataset-specific input resolutions are adopted. For DOTA-v1.0, the original high-resolution images are cropped into 1024×1024 patches with an overlap of 200 pixels. For HRSC2016 and DIOR, the input resolution is set to 800×800. These resolution settings are maintained during both training and testing. During training, random data augmentations are applied, each with a probability of 0.25. The images are then normalized using ImageNet statistics and padded with a size divisor of 32. During testing, random augmentations are disabled, and MultiScaleFlipAug is used to organize the data transformations and inference pipeline at the corresponding input resolutions.

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TABLE I

COMPARISON WITH SOTA METHODS ON THE DOTA-V1.0 TEST SET USING THE MULTI-SCALE CROPPING STRATEGY. CATEGORIES: PL–PLANE; BD–BASEBALL DIAMOND; BR–BRIDGE; GTF–GROUND TRACK FIELD; SV–SMALL VEHICLE; LV–LARGE VEHICLE; SH–SHIP; TC–TENNIS COURT; BC–BASEBALL FIELD; ST–STORAGE TANK; SBF–SOCCER-BALL FIELD; RA–ROUNDABOUT; HA–HARBOR; SP–SWIMMING POOL; HC–HELICOPTER. PRE-TRAINING: IN–SUPERVISED PRE-TRAINING ON IMAGENET; CO–SUPERVISED PRE-TRAINING ON MS COCO; M†–SELF-SUPERVISED PRE-TRAINING ON MILLIONAID [36]. THE BACKBONE OF MSTENET IS PRE-TRAINED ON IMAGENET FOR 300 EPOCHS.

Method	Pre	Params	FLOPs	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
one-stage																			
R3Det [16]	IN	41.9M	336G	76.47	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.57	62.68	67.53	78.56	72.62
CFA [7]	IN	36.6M	194G	76.67	89.08	83.20	54.37	66.87	81.23	80.96	87.17	90.21	84.32	86.09	52.34	69.94	75.52	80.76	67.96
DAFNe [8]	IN	-	-	76.95	89.40	86.27	53.70	60.51	82.04	81.17	88.66	90.37	83.81	87.27	53.93	69.38	75.61	81.26	70.86
SASM [9]	IN	36.6M	194G	79.17	89.54	85.94	57.73	78.41	79.78	84.19	89.25	90.87	58.80	87.27	63.82	67.81	78.67	79.35	69.37
AO2-DETR [37]	IN	74.3M	304G	79.22	89.95	84.52	56.90	74.83	80.86	83.47	88.47	90.87	86.12	88.55	63.21	65.09	79.09	82.88	73.46
S^2ANet [21]	IN	38.6M	198G	79.42	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58
KLD [38]	IN	41.9M	336G	80.63	89.92	85.13	59.19	81.33	78.82	84.38	87.50	89.80	87.33	87.00	72.57	71.35	77.12	79.34	78.68
RTMDet-R [11]	CO	52.3M	205G	81.33	88.01	86.17	58.54	82.44	81.30	84.82	88.71	90.89	88.77	87.37	71.96	71.18	81.23	81.40	77.13
two-stage																			
SCRDet [25]	IN	-	-	72.61	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21
RoI Trans [19]	IN	55.1M	200G	74.61	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03
Gilding Vertex [39]	IN	41.1M	198G	75.02	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32
CenterMap [40]	IN	41.1M	198G	76.03	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06
CSL [41]	IN	37.4M	236G	76.17	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	69.60	68.04	73.83	71.10	68.93
ReDet [10]	IN	31.6M	-	80.10	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67
GWD [42]	IN	-	-	80.23	89.66	84.99	59.26	82.19	78.97	84.83	87.70	90.21	86.54	86.85	73.47	67.77	76.92	79.22	74.92
DEA [43]	IN	-	-	80.37	89.92	83.84	59.65	79.88	80.11	87.96	88.17	90.31	88.93	88.46	68.93	65.94	78.04	79.69	75.78
DODet [44]	IN	-	-	80.62	89.96	85.52	58.01	81.22	78.71	85.46	88.59	90.89	87.12	87.80	70.50	71.54	82.06	77.43	74.47
AOPG [15]	IN	-	-	80.66	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10
O-RCNN [23]	IN	41.1M	199G	80.87	89.84	85.43	61.09	79.82	79.71	85.35	88.82	90.88	86.68	87.73	72.21	70.80	82.42	78.18	74.11
KFIoU [45]	IN	58.8M	206G	80.93	89.44	84.41	62.22	82.51	80.10	86.07	88.68	90.90	87.32	88.38	72.80	71.95	78.96	74.95	75.27
RVSA-O [12]	M+	114.4M	414G	81.24	88.97	85.76	61.46	81.27	79.98	85.31	88.30	90.84	85.06	87.50	66.77	73.11	84.75	81.88	77.58
LSKNet [17]	IN	31.0M	161G	81.85	89.69	85.70	61.47	83.23	81.37	86.05	88.64	90.88	88.49	87.40	71.67	71.35	79.19	81.77	80.86
PKINet [18]	IN	30.8M	-	80.10	89.72	84.20	55.81	77.63	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	77.47	73.62	88.62
STD-O [46]	IN	-	-	82.24	89.15	85.03	60.79	82.06	80.90	85.76	88.45	90.83	87.71	87.29	73.99	71.25	85.18	82.17	82.95
MSTENet-T(ours)	**IN**	**30.1M**	**172G**	**81.59**	**90.21**	**84.27**	**64.37**	**80.32**	**77.85**	**87.04**	**80.75**	**90.81**	**89.97**	**71.94**	**74.11**	**77.54**	**78.63**	**87.22**	**89.27**
MSTENet-S(ours)	**IN**	**39.7M**	**209G**	**82.72**	**90.19**	**85.35**	**66.54**	**81.07**	**83.42**	**87.36**	**87.51**	**90.82**	**89.82**	**71.79**	**74.65**	**77.06**	**78.51**	**87.39**	**89.62**

TABLE II

COMPARISON WITH SOTA METHODS ON THE HRSC2016 DATASET. THE COMMONLY USED MAP(07) [47] AND MAP(12) [48] METRICS ON HRSC2016 ARE ADOPTED FOR EVALUATION.

Method	Pre	Params	FLOPs	mAP(07)	mAP(12)
DRN [49]	IN	-	-	-	92.70
CenterMap [40]	IN	41.1M	198G	-	92.80
RoI Trans [19]	IN	55.1M	200G	86.20	-
G.V. [39]	IN	41.1M	198G	88.20	-
R3Det [16]	IN	41.9M	336G	89.26	96.01
DAL [50]	IN	36.4M	216G	89.77	-
GWD [42]	IN	47.4M	456G	89.85	97.37
S^2ANet [21]	IN	38.6M	198G	90.17	95.01
AOPG [15]	IN	-	-	90.34	96.22
ReDet [10]	IN	31.6M	-	90.46	97.63
O-RCNN [23]	IN	41.1M	199G	90.50	97.60
RTMDet [11]	CO	52.3M	205G	90.60	97.10
LSKNet-S [17]	IN	31.0M	161G	90.65	98.46
STD-O [46]	VIT-B	-	-	90.67	98.55
PKINet-S [18]	IN	30.8M	-	90.70	98.54
MSTENet-S(ours)	**IN**	**39.7M**	**209G**	**90.67**	**98.52**

TABLE III

COMPARISON WITH SOTA METHODS ON THE DIOR DATASET

Method	Pre	Params	FLOPs	mAP
RetinaNet-O [51]	IN	-	-	57.55
Faster RCNN-O [3]	IN	41.1M	198G	59.54
TIOE-Det [52]	IN	41.1M	198G	61.98
O-RCNN [23]	IN	41.1M	198G	64.30
ARS-DETR [53]	IN	41.9M	198G	66.12
O-RepPoints [20]	IN	36.6M	-	66.71
DCFL [54]	IN	-	-	66.80
LSKNet-S [17]	IN	31M	161G	65.90
PKINet-S [18]	IN	30.8M	190G	67.03
MSTENet-S(ours)	**IN**	**39.7M**	**209G**	**68.84**

C. Main Results

1) Results on DOTA-v1.0: Table I presents a quantitative comparison between MSTENet and representative oriented object detection methods developed in recent years on the DOTA dataset. As can be observed, compared with the classical O-RCNN framework with a ResNet backbone, MSTENet achieves a 1.85% improvement in mAP while requiring fewer parameters and a comparable computational cost, demonstrating superior detection performance and model efficiency. Moreover, compared with STD, a competitive ViT-based method, MSTENet still improves the mAP by 0.48%, further validating its competitive performance in oriented object detection.

Fig. 3 further presents qualitative comparisons between MSTENet and other SOTA methods on the DOTA dataset. It can be observed that MSTENet exhibits more pronounced advantages in detecting small-scale and elongated targets. This improvement mainly benefits from the collaborative modeling of STM and FEM, together with the dynamic regulation and adaptive fusion of their feature contributions by GFF, enabling the network to more accurately localize regions of interest and enhance the representation of targets at different scales and their fine-grained features. Even when a large number of targets with substantial scale variations coexist within the same image, MSTENet can adaptively balance the contributions of

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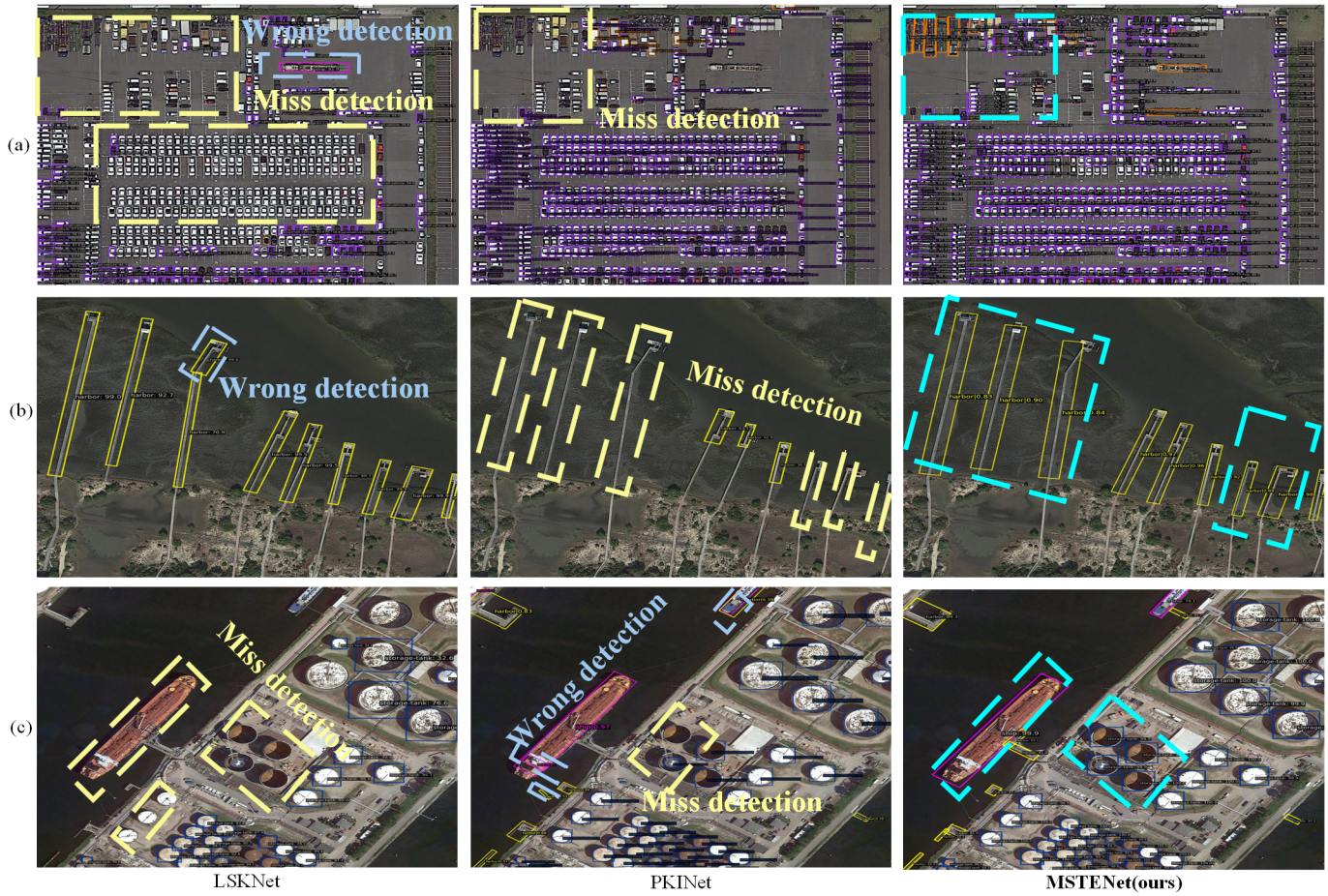

Fig. 3. Visual comparison of different detection frameworks on the DOTA-v1.0 dataset. (a) Detection results for small objects; (b) detection results for elongated objects; and (c) detection results for multi-scale objects.

STM and FEM according to the scale and structural characteristics of the targets, thereby effectively accommodating the detection requirements of targets at different scales.

2) Results on HRSC2012016: As shown in Table II, MSTENet achieves mAP (07) and mAP (12) scores of 90.67% and 98.52%, respectively, outperforming all compared methods except PKINet. The competitive performance demonstrates the effectiveness of MSTENet in extracting discriminative features and accurately detecting oriented ship targets. It should be noted that HRSC2016 predominantly consists of large warships and contains only a single object category, resulting in relatively limited variations in object scale and appearance. Therefore, although MSTENet achieves highly competitive results on this dataset, HRSC2016 does not fully exploit the capability of the proposed method in handling objects with substantial scale variations. The advantages of MSTENet in multi-scale feature representation can be more comprehensively evaluated on datasets with more diverse object categories and scale distributions.

3) Results on DIOR: As shown in Table III, MSTENet achieves an mAP of 68.84% on the DIOR dataset, outperforming all existing methods. DIOR contains diverse object categories and complex remote sensing scenes, with considerable variations in object scale, appearance, and spatial distribution, which place high demands on the feature representation capability of the detector. Moreover, objects within the same category may exhibit substantial differences in size and visual characteristics under different imaging conditions, while objects from different categories may present similar appearance patterns. These characteristics further increase the difficulty of accurate object localization and classification. The superior performance of MSTENet demonstrates its effectiveness in capturing discriminative features of objects across different scales and maintaining robust detection performance under diverse and challenging scene conditions.

D. Ablation Study

To verify the rationality and effectiveness of the proposed MSTENet, detailed experimental analyses and ablation studies are conducted on the designed modules. All experiments are performed on the DOTA-v1.0 dataset and evaluated on the trainval set.

1) Effect of Scale Transformation Module: To investigate the effects of convolutional kernel size and stride on the feature extraction performance of STM, we conduct experiments with different kernel-size and stride configurations, as shown in Table IV. It can be observed that using single-scale convolutions, such as (3×3) or (11×11), fails to achieve the

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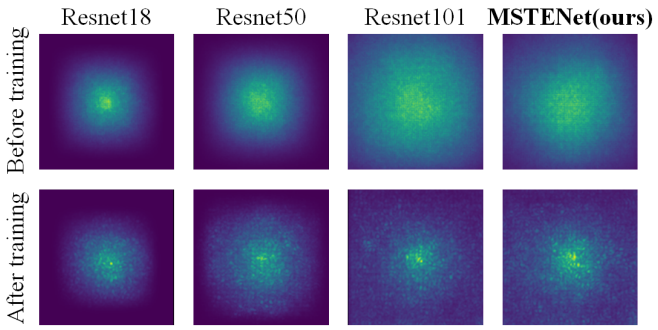

Fig. 4. Comparison of the ERFs of different models. Darker regions with broader spatial distributions indicate larger ERFs. In comparison, MSTENet exhibits a larger ERF.

Fig. 5. Receptive field activation maps of target objects. The white regions indicate the areas selected by FEM, while the bounding boxes denote the ground-truth locations of the objects. It can be observed that the regions selected by the network are consistently broader than the corresponding ground-truth bounding boxes, allowing FEM to capture sufficient surrounding background information and thereby enabling more accurate object category prediction.

optimal performance. In contrast, the multi-kernel structure exploits convolutional kernels of different sizes to capture multi-scale texture information, while incorporating grouped strip convolutions to enhance the representation of elongated targets with different orientations. Further experiments on the stride demonstrate that excessively large strides lead to the loss of fine-grained details, thereby degrading feature extraction performance. Considering the overall performance, we finally adopt a multi-scale combination of (3×3), (5×5), and (7×7) convolutional kernels.

TABLE IV
THE INFLUENCE OF THE SIZE AND STEP SIZE OF THE CONVOLUTIONAL KERNEL IN THE STM ON THE FEATURE EXTRACTION OF THE FEATURE BACKBONE.

Kernel Design	Params	FLOPs	mAP
(3×3) (3×3) (3×3)	3.25M	10.1G	75.64
(3×3) $(1\times5)\rightarrow(5\times1)\rightarrow(5\times5)$ $(7\times1)\rightarrow(1\times7)\rightarrow(7\times7)$	**4.27M**	**14.5G**	**78.23**
(3×3) $(1\times7)\longrightarrow(7\times1)\longrightarrow(7\times7)$ $(11\times1)\longrightarrow(1\times11)\longrightarrow(11\times11)$	4.44M	15.7G	77.89
(11×11) (11×11) (11×11)	5.46M	18.4G	78.11

To further analyze the receptive field characteristics of MSTENet, we visualize and compare its effective receptive field (ERF) with those of ResNets of different depths, as shown in Fig. 4. It can be observed that, before training, the receptive field gradually expands as the network depth increases. As training progresses, however, the receptive field is increasingly influenced by the convolutional structure. Compared with ResNet, MSTENet consistently exhibits a larger ERF at different stages, indicating that the scale transformation module can effectively enlarge the network receptive field and enhance its ability to capture contextual information of multi-scale objects.

2) Effect of Fine-grained Enhancement Module: To analyze the effect of strip convolution kernel size in FEM on feature extraction performance, we conduct comparative experiments with strip convolution kernels of different sizes, as shown in Table V. It can be observed that smaller convolution kernels have a limited receptive field and are therefore unable to fully exploit the contextual information surrounding the targets. Appropriately increasing the kernel size enlarges the feature perception range, thereby leading to better detection performance. Notably, the largest strip convolution kernel in the multi-scale kernel structure is (1×7), whereas FEM employs a larger (1×11) kernel. This is because, for large-scale objects, relying solely on the features of the objects themselves may lead to confusion between visually similar categories, such as bridges and harbors, or ships and vehicles. Therefore, FEM employs (1×11) strip convolutions to incorporate more comprehensive background contextual information, thereby enhancing the classification and localization capabilities of the network.

TABLE V
COMPARISON OF STRIP CONVOLUTION KERNEL SIZES IN THE FEM

Kernel Design	Params	FLOPs	mAP
$(1\times5)\longrightarrow(5\times1)$	1.32M	6.13G	78.34
$(1\times7)\longrightarrow(7\times1)$	1.63M	6.55G	78.77
$(1\times9)\longrightarrow(9\times1)$	1.94M	6.84G	79.11
$(1\times11)\longrightarrow(11\times1)$	**2.02M**	**7.08G**	**79.51**

Fig. 5 further illustrates the receptive field activation results of FEM for objects from different categories. It can be observed that the activation regions of FEM extend beyond the ground-truth object boundaries, indicating that the network can perceive broader surrounding background information while focusing on the intrinsic features of the objects. Such an enlarged contextual perception range enables the network to better exploit the correlations between objects and their surrounding scenes, thereby improving the discriminative capability for easily confused categories and enhancing object localization accuracy.

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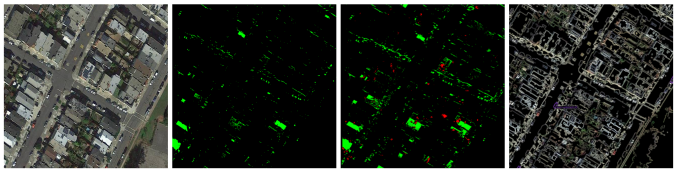

Fig. 6. Visualization of foreground–background information partitioning. (a) Ground Truth; (b) foreground–background information segmentation map in the spatial dimension; (c) foreground–background information segmentation map jointly modeled in the spatial and channel dimensions; (d) foreground–background information prediction map. Black denotes the background, green represents the information identified in the spatial dimension, and red represents the information identified in the channel dimension.

3) Effect of Gated Feature Fusion: To analyze the effects of information from different dimensions on the performance of gated feature fusion, we conduct ablation experiments on different feature mapping strategies along the channel and spatial dimensions, as shown in Table VI. It can be observed that both max pooling and average pooling along the channel dimension achieve favorable performance, and their combination further improves the mAP to 81.31%. In contrast, when only the spatial dimension is used for feature modeling, the mAP decreases to 65.27%, indicating that spatial information can reflect the positional distribution of objects but is insufficient to fully capture discriminative features across channels. By further jointly modeling the channel and spatial dimensions, the mAP reaches the highest value of 81.66%, demonstrating their strong complementarity. This enables GFF to measure feature importance from different dimensions and achieve more effective dynamic feature fusion.

TABLE VI
ABLATION STUDY ON WEIGHT ALLOCATION IN THE CHANNEL AND SPATIAL DIMENSIONS.

Channel		Spatial	Params	FLOPs	FPS	mAP
Max	Avg					
✓			20.63M	70.94G	20.7	81.23
	✓		20.61M	70.94G	20.7	81.12
✓	✓		20.65M	70.98G	20.7	81.31
		✓	20.13M	65.06G	17.3	65.21
✓	✓	✓	20.65M	71.10G	20.7	**81.66**

To further investigate the ability of gated feature fusion to model foreground and background information, we visualize the feature responses from different dimensions, as shown in Fig. 6. It can be observed that when only the spatial dimension is utilized, the model can localize the primary object regions, but some object responses remain incomplete. After incorporating the channel dimension, additional discriminative object information can be captured, resulting in a more complete representation of the foreground regions. The prediction results obtained by jointly modeling both dimensions effectively enhance the network's ability to distinguish foreground from background, thereby providing more effective guidance for feature fusion.

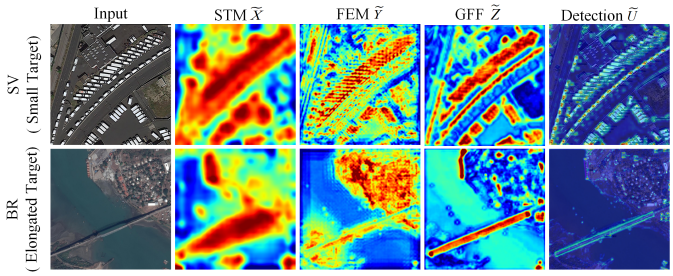

Fig. 7. Visualization of feature responses of different components in MSTENet. Small and elongated objects are selected as representative cases to compare the feature responses of STM, FEM, GFF, and the detection stage.

4) Ablation on MSTENet: To analyze the effects of STM, FEM, and GFF on model performance, we conduct ablation experiments, as shown in Table VII. When STM is used alone, the model achieves an mAP of 78.39%, whereas using FEM alone results in a significant performance degradation. This indicates that FEM relies on the multi-scale features provided by STM for fine-grained feature enhancement and cannot fully exploit its capability when used independently. Combining STM with FEM improves the mAP to 79.66%. With the further incorporation of GFF, the mAP increases to 80.78%, while the mRecall reaches 92.47%. Specifically, the AP/Recall of BR are improved by 0.80%/0.23%, respectively, while those of SV are improved by 0.83%/1.35%. These results demonstrate that GFF can effectively coordinate the feature contributions of STM and FEM, thereby improving the detection performance for objects at different scales.

TABLE VII
ABLATION STUDY OF DIFFERENT COMPONENTS IN MSTENET.

STM	FEM	GFF	BR		SV		mAP	mRecall
			AP	Recall	AP	Recall		
✓			77.89	85.63	78.97	84.88	78.39	91.77
	✓		59.97	66.82	66.73	77.17	55.46	77.54
✓	✓		78.11	86.10	78.56	87.12	79.66	92.05
✓	✓	✓	78.91	86.33	79.39	88.47	80.78	92.47

To further investigate the effects of different modules on feature representation, we visualize the feature responses at different stages, as shown in Fig. 7. After processing by STM, both the main body and elongated structures of the objects exhibit strong responses, although the feature distribution remains relatively scattered. Based on the output of STM, FEM further enhances object contours and local details, complementing and reinforcing the discriminative information preserved by STM. After GFF fusion, the object responses become more concentrated, providing clearer detail representations while maintaining the continuity of the overall object structure.

E. Extensions

1) Recall Rate Evaluation: In remote sensing scenarios, recall is also an important metric for evaluating model performance. To further verify the detection capability of MSTENet,

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TABLE VIII
COMPARISON OF RECALL RATES BY CATEGORY BETWEEN MSTENET AND SOTA METHODS ON THE DOTA DATASET.

Method	Recall														
	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
STD [46]	95.6	97.9	85.2	97.4	90.4	94.2	93.4	98.8	98.9	74.3	96.6	94.3	94.1	89.3	94.3
PKINet [18]	96.3	99.2	83.4	97.1	92.3	95.3	92.9	98.9	99.3	70.4	94.3	94.2	93.4	93.3	92.3
LSKNet [17]	95.6	98.0	89.1	97.0	90.4	94.6	93.4	99.1	99.1	71.3	94.3	94.3	93.1	91.3	93.4
MSTENet(ours)	96.4	99.1	**89.3**	97.2	**93.4**	95.6	94.3	99.0	99.1	75.2	96.4	95.6	94.3	93.4	95.6

we compare it with several state-of-the-art methods, as shown in Table VIII. It can be observed that MSTENet achieves superior overall recall, with particularly notable advantages for the BR and SV categories. For BR objects, which typically exhibit elongated structures with large spatial extents, STM captures more complete long-range structural information, while FEM further enhances edge and local detail features, thereby reducing fragmented responses and local omissions. For small-scale SV objects that are easily affected by background interference, FEM strengthens fine-grained feature representation, STM supplements multi-scale contextual information, and GFF adaptively coordinates the contributions of the two types of features, thereby effectively reducing the missed detection rate.

2) Analysis of the interpretability of internal mechanisms: To further enhance the transparency of the underlying mechanisms of MSTENet, we analyze its internal behavior from three perspectives: feature interpretability, decision pathways, and gradient attribution, aiming to reveal how the model performs feature representation, fusion decision-making, and final prediction.

TABLE IX
LOCATION FOR IMPLEMENTING STM AND FEM

Stage	Params	FLOPs	FPS	mAP
0	7.33M	49.48G	11.0	80.10
1	11.06M	53.21G	10.2	80.45
2	14.63M	56.78G	9.6	80.76
3	17.28M	67.64G	8.0	81.03
4	**20.65M**	**71.1G**	7.7	**81.11**

a) Feature interpretability: To analyze the roles of STM and FEM in feature extraction at different hierarchical levels, we progressively embed them into the four stages of the backbone network, with the results shown in Table IX. It can be observed that the model performance consistently improves as the modules are introduced stage by stage, demonstrating their effectiveness in enhancing multi-level feature extraction. In particular, deeper integration of STM and FEM allows the network to capture more complementary spatial and contextual information across different feature scales. When they are deployed in all four stages, the mAP increases by 1.3%, indicating that STM and FEM can continuously strengthen feature representation across different hierarchical levels and contribute to more accurate object detection.

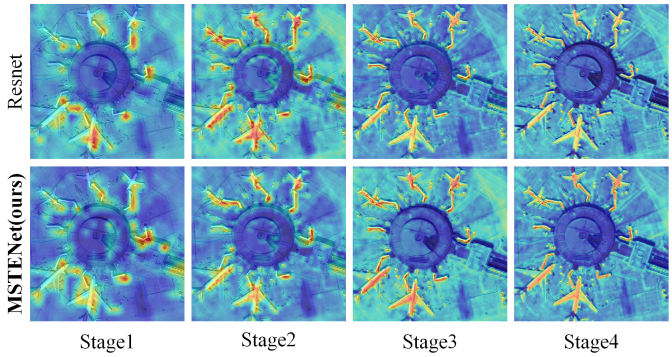

Fig. 8. Visual comparison of feature maps at different stages of various frameworks. Red represents a higher attention value, and blue represents a lower attention value.

Furthermore, Fig. 8 presents the feature responses at different stages, where MSTENet is compared with ResNet at the corresponding stages. Compared with ResNet, MSTENet highlights foreground objects more clearly and effectively suppresses irrelevant background responses at different stages, demonstrating that the proposed modules can progressively enhance the representation of discriminative object features.

b) Decision path: To investigate the effect of the gating parameter m on feature fusion in GFF, we select samples from different scenes and compare their feature responses under fixed and adaptive gating settings, as shown in Fig. 9. As m increases, the feature responses gradually shift from local details to broader regional information. However, excessively large values of m may introduce irrelevant background responses. For the first sample, GFF adaptively selects a relatively small value of $m = 0.32$, enhancing target responses while preserving object boundaries and local structural details. In contrast, the second sample contains densely distributed small vehicles and complex backgrounds, where smaller values of m tend to produce redundant local responses. The adaptively selected value of $m = 0.81$ effectively suppresses background interference and highlights target regions. These observations demonstrate that GFF does not rely on a fixed fusion ratio but dynamically balances the contributions of FEM and STM according to the feature requirements of different scenes.

Furthermore, we analyze the distribution of object scales in the DOTA dataset and the corresponding values of m

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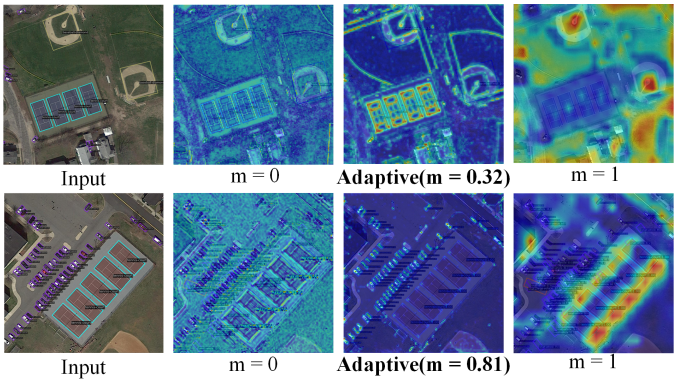

Fig. 9. Visualization of feature responses under different gating parameter m settings across different scenes.

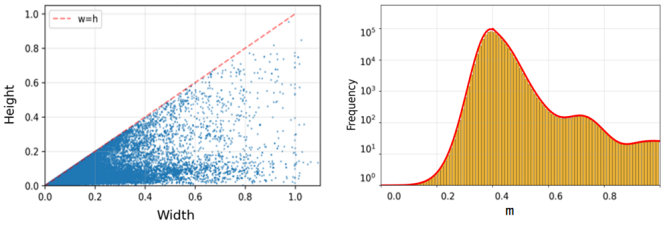

Fig. 10. Statistical analysis of target scale distribution and the adaptive parameter m in the DOTA dataset. The left figure shows that most target sizes are concentrated near the origin, indicating that the dataset contains a large number of small objects. The right figure shows that the distribution of the corresponding adaptive parameter m is generally biased toward smaller values, suggesting that the FEM module plays a dominant role. This is consistent with the characteristic that small objects account for a high proportion in the DOTA dataset.

learned by the model, as shown in Fig. 10(a) and (b), respectively. It can be observed that most objects in DOTA are concentrated in the small-scale range, while the learned values of m are predominantly distributed in the lower range. This indicates that GFF generally places greater emphasis on FEM to preserve fine-grained information on DOTA, while adaptively increasing the contribution of STM for specific scenes, consistent with its design objective of dynamically balancing the two feature branches.

c) gradient attribution: In terms of gradient analysis, we further investigate the model decision-making process from three perspectives: branch sensitivity, target-region concentration, and gating sensitivity, with the results shown in Table X.

TABLE X
GRADIENT ATTRIBUTION ANALYSIS OF MSTENET FOR DIFFERENT TARGET TYPES.

Target type	Branch sensitivity		Target localization		Gate sensitivity	
	G_{STM}	G_{FEM}	L_{STM}	L_{FEM}	G_M	G_{GFF}
SV (small target)	0.47	0.71	0.64	0.82	0.68	0.77
BR (elongated target)	0.76	0.53	0.86	0.67	0.41	0.81

For small-scale objects such as SV, FEM exhibits higher gradient sensitivity G_{FEM} and target-region concentration L_{FEM}, indicating that the model predictions rely more heavily on the fine-grained features provided by FEM and that the

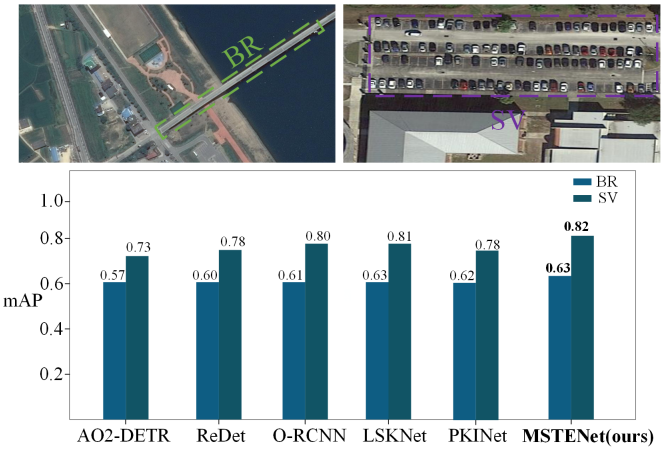

Fig. 11. Comparison of Elongated objects and small target objects in different detection frameworks.

corresponding gradients are more effectively concentrated on the target regions. In contrast, for objects with pronounced elongated structures, such as BR, STM achieves higher G_{STM} and L_{STM}, suggesting that structural representation plays a more important role in recognizing such objects. In addition, both categories exhibit evident gating sensitivity G_M, demonstrating that variations in the gating variable m have a direct influence on the final predictions rather than merely serving as an intermediate fusion parameter. The fused GFF features further maintain stable gradient responses. These results are consistent with the adaptive variation trend of the gating parameter discussed above, further validating the dynamic feature coordination capability of GFF from the gradient perspective.

3) Analysis of Small and Elongated Targets: To further reveal the feature adaptation mechanism of MSTENet for small and elongated objects, we compare the detection results of different detection frameworks, as shown in Fig. 11. Compared with LSKNet and PKINet, MSTENet produces more complete responses for vehicle targets and preserves the structures of elongated objects, such as bridges, more consistently. These differences are closely related to the distinct feature representation requirements of the two types of objects. Small objects have limited spatial extents, and their detailed information is easily lost during feature extraction, making them more dependent on local textures and neighborhood context. In contrast, elongated objects exhibit pronounced directional characteristics and large spatial spans, and therefore rely more heavily on long-range structural information.

To address these characteristics, FEM enhances fine-grained features and incorporates neighborhood information to improve the discriminability of small objects, while STM enlarges the receptive field through multi-scale convolutions and directional strip convolutions, thereby enhancing structural continuity along the extending directions of elongated objects. On this basis, GFF adaptively coordinates the feature contributions of STM and FEM, enabling the network to select more appropriate feature representations according to different object morphologies. As a result, MSTENet can simultane-

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ously preserve the detailed information of small objects and effectively model the structures of elongated objects.

4) Adaptability to Different Backbones: To verify the superior compatibility of MSTENet, we replace the backbones of different detection frameworks with MSTENet. As shown in Table XI, MSTENet can be effectively integrated into both two-stage detectors, including ORCNN and RoI Transformer, and one-stage detectors, including S2A-Net and R3Det. Compared with ResNet-18 and LSKNet used as backbones in these frameworks, MSTENet requires substantially fewer parameters and FLOPs than ResNet-18 and only slightly more than LSKNet, while achieving significantly better detection performance than both. These results demonstrate the strong compatibility and effectiveness of MSTENet across different detection frameworks.

TABLE XI
COMPARISON OF DETECTION PERFORMANCE AND MODEL COMPLEXITY WITH THE BACKBONE NETWORK.

Detector	ResNet-18	LSKNet	**MSTENet (ours)**
O-RCNN [23]	79.27	81.31	**82.64**
RoI Trans [19]	78.32	80.89	79.80
S^2ANet [21]	76.82	80.15	**81.72**
R3Det [10]	74.16	78.39	**80.55**
Params	11.2M	4.3M	6.4M
FLOPs	38.1G	19.1G	24.7G

5) Training Efficiency: To comprehensively evaluate the efficiency of different detectors, we conduct experiments under a unified configuration, with the results shown in Table XII. All models are evaluated on an NVIDIA L20 GPU with a batch size of 4 and follow the default training schedules of the corresponding methods. Specifically, O-RCNN adopts the official $2\times$ training schedule for sufficient optimization, resulting in a higher training cost, whereas MSTENet achieves strong detection performance with only a $1\times$ schedule. Meanwhile, O-RCNN employs the conventional ResNet-50 backbone, while MSTENet, LSKNet, and PKINet adopt lightweight backbones specifically designed for remote sensing scenarios, which helps reduce the computational overhead of feature extraction and maintain relatively high inference efficiency.

TABLE XII
EFFICIENCY COMPARISON OF DIFFERENT DETECTORS

Method	mAP	Schedule	FPS	Training (h)
O-RCNN [23]	80.87	2x	15.1	15
LSKNet [17]	81.85	1x	21.9	9
PKINet [18]	80.10	1x	**23.4**	**8**
MSTENet(ours)	**82.72**	1x	21.5	9

Compared with LSKNet, MSTENet improves the mAP from 81.85% to 82.72%, with only a slight decrease in FPS from 21.9 to 21.5, indicating that the proposed modules achieve a clear accuracy gain with limited additional computational overhead. Although PKINet achieves the highest inference speed of 23.4 FPS, its mAP is only 80.10%. O-RCNN achieves only 15.1 FPS and also yields lower detection accuracy than MSTENet. Overall, MSTENet requires only about 9 h of training while maintaining both high detection accuracy and competitive inference efficiency, demonstrating a better trade-off among accuracy, inference speed, and training cost.

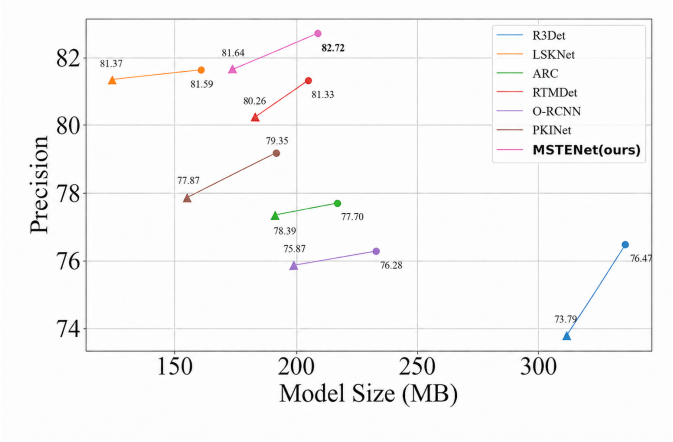

Fig. 12. Comparison of remote sensing detection frameworks with different model sizes. The ▲ and ● denote the Tiny and Small configurations, respectively. It can be observed that MSTENet outperforms all the mainstream models considered.

Meanwhile, to further evaluate the overall performance of different detection frameworks in terms of detection accuracy and model size, we visualize and compare representative methods on the DOTA dataset, as shown in Fig. 12. It can be observed that MSTENet achieves higher detection accuracy under both the Tiny and Small configurations without significantly increasing the model size. Compared with other methods, MSTENet exhibits a more pronounced accuracy advantage at a comparable model size, indicating that it can achieve stronger feature representation capability with relatively low complexity and attain a superior balance between detection performance and model size.

V. CONCLUSION

In this paper, we propose MSTENet for remote sensing object detection to address large object-scale variations and foreground-background ambiguity. STM captures multi-scale contextual information with enlarged receptive fields, FEM enhances fine-grained local details, and GFF adaptively integrates global and local features for more effective representation. MSTENet achieves 82.72%, 90.67%, and 68.84% on DOTA, HRSC2016, and DIOR, respectively, outperforming state-of-the-art methods by 0.5% and 1.8% on DOTA and DIOR while achieving comparable performance on HRSC2016.

MSTENet still has limitations in densely distributed, occluded, or strongly cluttered scenes, where noisy features may reduce fusion stability. In addition, its multi-module design introduces extra computational overhead. Future work will focus on more robust foreground-background modeling and lightweight feature fusion, together with model compression, to improve robustness and deployment efficiency.

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